Final Report

AuraBot – Hand Gesture Controlled Robot using NodeMCU + MPU6050

1. Introduction

AuraBot is a wireless robot that moves according to the gestures of the human hand. It uses an MPU6050 accelerometer–gyroscope sensor mounted on a glove to capture hand motion and sends data to another NodeMCU placed on the robot body. The project combines IoT, robotics, and sensor fusion concepts to demonstrate human–machine interaction.

2. Objectives

- To design and implement a wireless hand gesture-controlled robot.
- To understand the working principles of the MPU6050 sensor and NodeMCU communication.
- To develop a simple and efficient control algorithm for gesture detection.
- To enhance hands-on skills in electronics, sensors, and embedded programming.
- To test and analyze the system's response and accuracy to various gestures.

3. Components Used

Breadboard

MPU6050 Accelerometer Module

NodeMCU (ESP8266)

H-Bridge

DC Motors

Wheels and caster wheel

Battery

Cables As needed

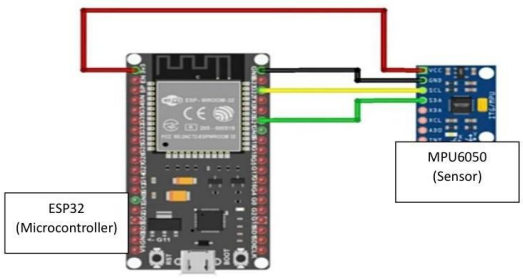
4. Working Principle

The MPU6050 sensor detects the orientation of the hand in terms of X, Y, and Z axes. The sensor data is transmitted wirelessly...
Based on the tilt direction of the hand:
- Forward Tilt → Robot moves forward
- Backward Tilt → Robot moves backward
- Left Tilt → Robot turns left
- Right Tilt → Robot turns right
- Hand Flat → Robot stops

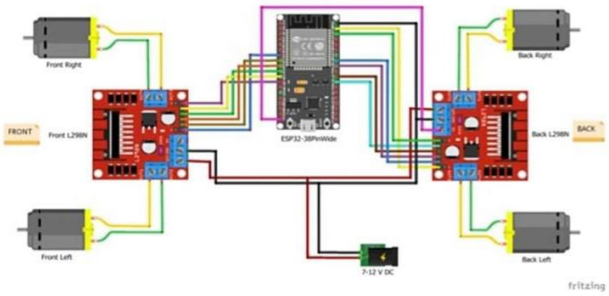
5. Circuit Description

The system consists of two circuits:
- Transmitter Circuit (Glove Unit)
- Receiver Circuit (Robot Unit)

Circuit Schematic:



Transmitter Circuit



Receiver Circuit

6. Software and Control

Implemented using Arduino IDE. Includes initialization, gesture reading, and Wi-Fi communication.

```
#include <ESP8266WiFi.h>
#include <WiFiUdp.h>

const char* SSID = "ROBOT_AP";
const char* PASS = "12345678";
const uint16_t PORT = 4210;

const int IN1 = D5;
const int IN2 = D6;
const int IN3 = D7;
const int IN4 = D8;
const int ENA = D3;
const int ENB = D4;

const int PWM_MAX = 1023, BASE_DUTY = 650, MAX_DUTY = 950;
const float DEAD_DEG = 6.0f, MAX_DEG = 25.0f;
const unsigned long STILL_TIMEOUT_MS = 800;
const float STILL_ANGLE_DELTA = 1.5f;

WiFiUDP udp;
float lastRoll = 0, lastPitch = 0;
unsigned long lastMoveMs = 0;

inline int iconstrain(int v, int lo, int hi){ return v < lo ? lo : (v > hi ? hi : v); }
inline float fconstrain(float v, float lo, float hi){ return v < lo ? lo : (v > hi ? hi : v); }

void fwdA(int d){ analogWrite(IN1, d); analogWrite(IN2, 0); }
void bwdA(int d){ analogWrite(IN1, 0); analogWrite(IN2, d); }
void stpA(){ analogWrite(IN1, 0); analogWrite(IN2, 0); }

void fwdB(int d){ analogWrite(IN3, d); analogWrite(IN4, 0); }
void bwdB(int d){ analogWrite(IN3, 0); analogWrite(IN4, d); }
void stpB(){ analogWrite(IN3, 0); analogWrite(IN4, 0); }
```

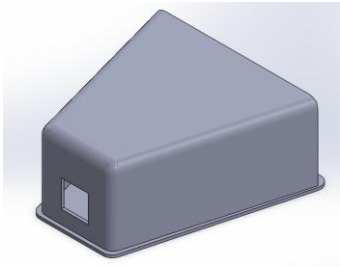
part of the Receiver's Code
(the Esp placed in the cars body)

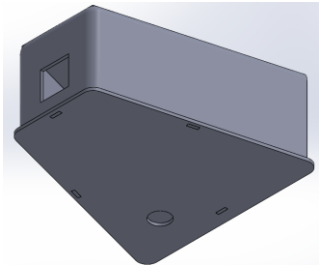
```
bool i2cW(uint8_t d,uint8_t r,uint8_t v){ Wire.beginTransmission(d); Wire.write(r); Wire.write(v); return Wire.endTransmission()==0; }
bool i2cR(uint8_t d,uint8_t r,uint8_t* b,uint8_t n){ Wire.beginTransmission(d); Wire.write(r); if(Wire.endTransmission(false)!=0) return false; uint8_t m=Wire.requestFro
bool det(){ for(uint8_t a:{MPU_ADDR_LOW,MPU_ADDR_HIGH}){ uint8_t w=0; if(i2cR(a,REG_WHO_AM_I,&w,1)&&w==0x68){ MPU_ADDR=a; return true; } } return false; }
bool initIMU(){ if(!i2cW(MPU_ADDR,REG_PWR_MGMT_1,0x00)) return false; delay(30); return i2cW(MPU_ADDR,REG_SMPLRT_DIV,0x07)&&i2cW(MPU_ADDR,REG_CONFIG,0x03)&&i2cW(MPU_ADDR
bool readAcc(int16_t& ax,int16_t& ay,int16_t& az){ uint8_t b[6]; if(!i2cR(MPU_ADDR,REG_ACCEL_XOUT_H,b,6)) return false; ax=(int16_t)((b[0]<<8)|b[1]); ay=(int16_t)((b[2]<

void setup(){
  Serial.begin(115200);
  Wire.begin(D2,D1);
  det(); initIMU();
  const int N=300; long sax=0,say=0,saz=0;
  for(int i=0;i<N;i++){ int16_t ax,ay,az; if(readAcc(ax,ay,az)){ sax+=ax; say+=ay; saz+=az; } delay(2); }
  ax_off=(float)sax/N; ay_off=(float)say/N; az_off=(float)saz/N-ACC_SENS;

  WiFi.mode(WIFI_STA);
  WiFi.begin(SSID,PASS);
  while (WiFi.status()!=WL_CONNECTED) { delay(200); }
  udp.begin(PORT);
  Serial.println("Glove sender ready");
}

void loop(){
  int16_t axr,ayr,azr;
  if(!readAcc(axr,ayr,azr)){ delay(5); return; }
  float ax=(axr-ax_off)/ACC_SENS;
  float ay=(ayr-ay_off)/ACC_SENS;
  float az=(azr-az_off)/ACC_SENS;
  float roll = atan2f(ay,az)*180.0f/PI;
  float pitch= atan2f(-ax, sqrtf(ay*ay+az*az))*180.0f/PI;

  char buf[64];
  int n = snprintf(buf,sizeof(buf),"%.2f,%.2f",roll,pitch);
  udp.beginPacket(HOST_IP,PORT);
```

A part of The Transmitter's Code
(the ESP placed in the glove).

7. Mechanical Design

The car's chassis was fully designed and fabricated using 3D printing and interlocking joints, combining hardware design, motion sensing, wireless data transfer, and control systems into one cohesive setup.





8.

Testing and Results

Gesture	Expected Motion	Observed Motion	Accuracy
Forward tilt	Move Forward	✓	98%
Backward tilt	Move Backward	✓	96%
Left tilt	Turn Left	✓	95%
Right tilt	Turn Right	✓	94%
Flat hand	Stop	✓	100%

9. Conclusion

AuraBot demonstrates how human gestures can intuitively control robotic systems. The prototype successfully integrates motion sensing and wireless communication.

10. Recommendations

- Integrate Bluetooth or RF modules for longer range.
- Add obstacle avoidance sensors.
- Improve mechanical design.
- Use machine learning for gesture classification.